

Project Synopsis

on

Smart Video Conferencing Application

In the partial fulfilment for the award of

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IN

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Index

1. Introduction	1
1.1. Evolution of Video Conferencing	1
1.2. Limitations of Current Systems	1
1.3. Role of Artificial Intelligence and Analytics	1
1.4. Proposed Project Approach	2
1.5. Significance of the Project	2
2. Proposed Work	3
3. Objectives	4
4. Methodology to be used	5
4.1. Proposed Methodology	5
4.2. Proposed Technology	6
5. Hardware Software requirements	7
5.1. Hardware requirements	7
5.2. Software requirements	7
References	8

In recent years, video conferencing has become a vital tool for online education, corporate meetings, remote collaboration, and virtual events. Platforms like Zoom, Google Meet, and Microsoft Teams have gained immense popularity due to their ability to connect people across geographical boundaries.

1.1. Evolution of Video Conferencing

Over the past decade, video conferencing has become one of the most widely used technologies in education, business, and day-to-day communication. What once started as a luxury tool for corporate boardrooms is now a necessity for schools, universities, offices, and even small groups of people who want to collaborate virtually. Especially after the COVID-19 pandemic, the reliance on online meetings has grown dramatically, with platforms such as Zoom, Google Meet, and Microsoft Teams becoming household names. These applications made it possible to connect people across the world with just a stable internet connection.

1.2. Limitations of Current Systems

While these tools are excellent for real-time communication, most of them stop at providing only the basic features—video, audio, screen sharing, and a chat box. As virtual meetings replaced physical ones, several challenges came into focus. A teacher in a physical classroom can instantly sense if students are attentive, distracted, or even absent, but the same is not possible in an online class where cameras are turned off or where participants can easily get distracted by other activities. Similarly, in professional meetings, managers or team leaders cannot always judge how engaged their team members are, making it harder to measure participation or productivity.

Another major issue is **attendance tracking**. In most current systems, attendance is recorded manually or by checking who joined and left the meeting. This does not reveal whether participants were genuinely present and attentive during the session. On top of that, preparing minutes of the meeting (MoM) is still a manual and time-consuming task. Hosts or coordinators often have to spend additional time after a meeting summarizing the discussion, which is inefficient and prone to missing important points.

1.3. Role of Artificial Intelligence and Analytics

These challenges open up an opportunity to make online meetings smarter with the help of **Artificial Intelligence (AI)** and **data analytics**. AI has already shown its potential in fields like image recognition, natural language processing, and predictive analysis. If applied to video conferencing, it can solve some of the problems faced by both educators and professionals.

For example:

- **AI-based attentiveness detection** can monitor participants through facial expressions, eye movements, and posture analysis.
- **Speech-to-text technology** can automatically transcribe spoken content.

- **NLP summarization models** can generate concise and meaningful minutes of meeting.
- **Data analytics** can provide insights such as how many meetings a student has attended, their overall attentiveness percentage, or how engaged a team was in a particular session.

1.4. Proposed Project Approach

The proposed project, therefore, aims to develop a **Smart AI-Powered Video Conferencing Application** that not only enables smooth communication but also integrates these advanced features. The system will include modules for:

- Real-time video conferencing.
- AI-based attentiveness detection.
- Automated meeting summarization.
- Attendance management.
- Personalized dashboards for both hosts and participants.

The host will be able to view analytics such as **engagement heatmaps, participation percentages, and attentiveness levels**, while each participant will have their own profile showing meeting history, attendance, and performance trends.

1.5. Significance of the Project

The significance of such a system is clear. In the field of **education**, teachers will be able to track not only whether students are present but also whether they are genuinely paying attention, allowing them to identify weaker students or improve teaching strategies. In **corporate environments**, managers can save time with automatically generated meeting summaries and gain better insights into employee participation. For **event organizers**, the system can provide statistics on how many attendees stayed throughout a session, which parts of the event had higher engagement, and overall feedback on effectiveness.

The proposed project aims to design and develop a **smart video conferencing platform** that not only enables smooth and secure online meetings but also integrates **artificial intelligence features** to enhance the overall experience for both hosts and participants. This system will focus on **attentiveness detection, automated meeting summaries, attendance tracking, and analytical insights** for participants and organizers.

The system will be divided into multiple interconnected modules:

a) Backend Services

The backend will be developed using **Node.js and Express.js** to handle user authentication, meeting creation, participant management, and interaction between different system components.

b) Video Conferencing Engine

Real-time communication will be enabled through **WebRTC**, which ensures smooth video and audio streaming. This engine will be responsible for handling screen sharing, chat, and live interactions during meetings.

c) AI & Machine Learning Modules

- **Computer Vision (CV):** To monitor participants' attentiveness using eye-tracking, facial expressions, and movement analysis.
- **Natural Language Processing (NLP):** To automatically generate meeting summaries and minutes after each session.
- **Speech Analytics:** To detect speaking time, keyword frequency, and engagement levels.

d) Database Management

MongoDB will be used to store user data, meeting records, attendance logs, and AI-generated reports securely.

e) Analytics & Insights

After every meeting, hosts and participants will get access to detailed reports, including:

- Attendance logs.
- Attentiveness scores and trends.
- Speaking time distribution.
- Meeting summaries and keyword highlights.
- Percentage of students or employees participating across sessions.

f) Integration of Dashboard

A dashboard will display visual insights such as **graphs, charts, and heatmaps** for better understanding of meeting engagement and performance trends.

The primary objectives of this project are to design and develop an **AI-powered engagement analysis and summarization system for online meetings**.

This project aims:

1. **To implement real-time engagement analytics** by tracking participants' eye movements and body posture to determine attentiveness during meetings.
2. **To Generate automatic meeting summaries** using speech-to-text transcription combined with natural language processing techniques, ensuring that key discussion points and action items are captured without manual note-taking.
3. **To Provide a user-friendly dashboard** to visualize engagement levels, attendance details, and summaries, enabling hosts and participants to evaluate meeting effectiveness.
4. **To Enhance productivity** in academic and professional environments by offering both qualitative (awareness, attentiveness) and quantitative (time spent, participation ratio) insights.

4.

Methodology/Technology to be Used

4.1. Proposed Methodology

a) Integration with Meeting SDK:

- A media server, such as Janus, will be implemented to handle the core functionalities of real-time video and audio streaming. Using WebRTC, client devices will establish direct peer connections, which are routed through Janus for mixing, recording, and distribution.
- This ensures a flexible, scalable infrastructure, allowing real-time processing of video data.

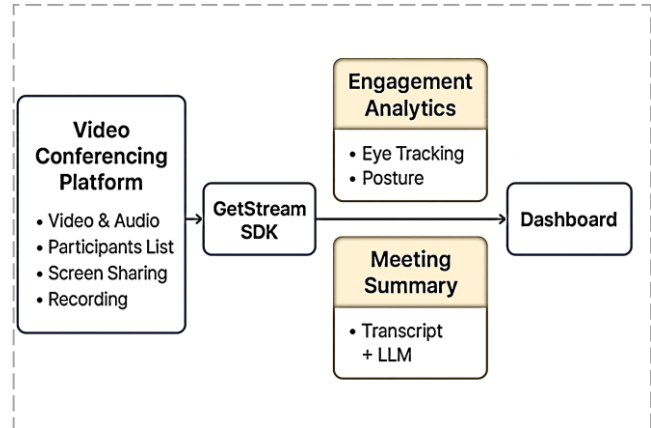


Fig.4.1.: Methodology

b) Engagement Analytics Module:

- Real-time video streams of participants will be accessed and processed to monitor **eye gaze direction** and **body posture**.
- Computer Vision models will be used for detecting attentiveness and posture deviations.
- The results will be aggregated into an **engagement score** for each participant.

c) Meeting Summarization Module:

- Audio streams will be transcribed using speech-to-text systems such as **OpenAI Whisper** or **Google Speech-to-Text API**.
- Natural Language Processing techniques (e.g., **transformer-based summarizers**) will generate concise summaries highlighting the key discussion points and action items.

d) Data Visualization and Reporting:

- A **React.js frontend** will display real-time engagement dashboards showing participant attentiveness, hand raises, and activity.
- Summaries and analytics will be stored in a backend database (**MongoDB**) for generating reports and maintaining meeting history.

e) Backend Processing and Storage:

- A backend server built with **Node.js** and **Express.js** will manage user sessions, handle API requests, and process heavy AI computations (if offloaded from the client).
- Meeting engagement reports and summaries will be securely stored in a database.

4.2. Proposed Technology

- **Frontend:** React.js / Next.js, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Video SDK:** WebRTC, Janus, Socket.IO
- **Computer Vision:** MediaPipe, OpenCV, TensorFlow.js (for eye tracking & posture detection)
- **Speech-to-Text & NLP:** OpenAI Whisper / Google STT, Transformer-based summarizers (e.g., BERT, GPT)
- **Visualization:** Custom React dashboard with charts & analytics

5.

Hardware Software Requirements

5.1. Hardware Requirements

To successfully implement and demonstrate the project, the following hardware resources are required:

- **Processor:** Intel i5 (10th Gen or above) / AMD Ryzen 5 or equivalent
- **RAM:** Minimum 8 GB (16 GB recommended for smooth AI model execution)
- **Storage:** At least 256 GB SSD (for faster data access and storing meeting recordings/summaries)
- **GPU (Optional but recommended):** NVIDIA GPU with CUDA support (for faster AI inference in eye-tracking and summarization modules)
- **Camera & Microphone:** Standard HD webcam and microphone for video/audio capture during meetings
- **Internet Connection:** Stable broadband connection (minimum 5 Mbps upload/download) for real-time communication.

5.2. Software Requirements

- **Operating System:** Windows 10/11 / Ubuntu Linux (development & testing)
- **MERN** to develop project's Frontend and Backend.
- **AI/ML Libraries** like OpenCV, TensorFlow, MediaPipe.
- **Video Conferencing SDKs** like WebRTC, Socket.IO, Twilio, (for video/audio meeting infrastructure)
- **Development Tools:**
 - Visual Studio Code / WebStorm (IDE)
 - Postman (API testing)
 - Git & GitHub (version control)
 - Docker (optional, for deployment and containerization)

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